Chromium Browser Artifact Parser

User's Guide

Version: 2025-May-20 (Enhanced UI Edition)
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1. Introduction

The Chromium Browser Artifact Parser is a forensic tool designed to extract and parse browser artifacts from Chromium-based web browsers. It provides a graphical user interface (GUI) that simplifies the process of extracting valuable forensic data from browser profiles and exports the results to a well-organized Microsoft Excel spreadsheet.

Key Features

- **Modern GUI Interface** - User-friendly graphical interface with progress tracking and status updates
- **Selective Artifact Processing** - Choose which artifacts to extract based on your investigation needs
- **Comprehensive Data Extraction** - Parses 20 different artifact categories from browser profiles
- **Multi-Browser Support** - Works with Chrome, Microsoft Edge, Opera, and other Chromium-based browsers

- **Gap Detection** - Identifies deleted records by detecting gaps in database record numbering
- **Timestamp Decoding** - Automatically converts WebKit timestamps to human-readable UTC format
- **Excel Output** - Organizes all extracted data into a multi-worksheet Excel file with a summary sheet

Use Cases

- Digital forensic investigations
- Incident response
- Employee monitoring (with proper authorization)
- Personal data recovery
- Security audits

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2. System Requirements

Hardware Requirements

- Minimum 4 GB RAM (8 GB recommended)
- 100 MB free disk space for the application
- Additional disk space for output files (varies based on browser data volume)

Software Requirements

- **Operating System:** Windows 10/11 (primary), also works on macOS and Linux
- **Python:** Version 3.10 or higher
- **Required Python Packages:**
  - pandas ~= 2.0.1
  - openpyxl ~= 3.1.5
  - numpy ~= 1.24.3
  - DateTime ~= 5.5
  - tkinter (typically included with Python)

Browser Compatibility

- Google Chrome (tested with versions 129 and 143)
- Microsoft Edge (tested with version 129)
- Opera (tested with version 113)

- Other Chromium-based browsers (Brave, Vivaldi, etc.)

Note: The tool is designed for desktop browser profiles. Mobile browser profiles may have different database structures and are not currently supported.

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3. Installation

Option 1: Running from Source (Recommended)

1. **Download the Repository** Clone or download the repository from GitHub:

```
https://github.com/jjrboucher/Chromium-Browser-Artifact-Parser
```

2. **Install Python Dependencies** Open a command prompt or terminal in the project directory and run:

```bash
pip install -r requirements.txt
```

3. **Verify Installation** Run the application:

```bash
python browser-artifact-parser-GUI.py
```

Option 2: Using the Standalone Executable (Windows Only)

1. Download `browser-artifact-parser-GUI.exe` from the repository
2. Double-click to run (no Python installation required)

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4. Getting Started

Locating Browser Profile Folders

Before using the parser, you need to locate the browser profile folder you want to analyze. Here are the default locations:

Google Chrome (Windows):

```
C:\Users\<username>\AppData\Local\Google\Chrome\User Data\Default
```

Google Chrome (macOS):

```
/Users/<username>/Library/Application Support/Google/Chrome/Default
```

Google Chrome (Linux):

```
/home/<username>/.config/google-chrome/Default
```

Microsoft Edge (Windows):

```
C:\Users\<username>\AppData\Local\Microsoft\Edge\User Data\Default
```

Opera (Windows):

```
C:\Users\<username>\AppData\Roaming\Opera Software\Opera Stable
```

Note: If the user has signed into the browser, profiles may be named `Profile 1`, `Profile 2`, etc., instead of `Default`.

Important Considerations

1. **Close the Browser First** - Browser databases are locked while the browser is running. Close all browser windows before parsing to avoid "database is locked" errors.
2. **Make a Copy** - For forensic investigations, always work with a copy of the profile folder, not the original.
3. **Administrator Access** - You may need administrator privileges to access profile folders in protected locations.

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5. Using the Application

Launching the Application

Run the application using one of these methods:

- **Python:** `python browser-artifact-parser-GUI.py`
- **Windows Executable:** Double-click `browser-artifact-parser-GUI.exe`

Main Interface Overview

The application window contains the following sections:

1. **Chrome User Profile Folder** - Path to the browser profile to analyze
2. **Output Excel File** - Path where the results will be saved
3. **Artifact Selection** - Checkboxes to select which artifacts to process
4. **Progress Section** - Progress bar and control buttons
5. **Processing Status** - Real-time log of processing activities

Step-by-Step Instructions

Step 1: Select Browser Profile Folder

1. Click the 📁 **Browse** button next to "Chrome User Profile Folder"
2. Navigate to and select the browser profile folder (e.g., `Default`)
3. The path will appear in the text field

Step 2: Select Output File Location

1. Click the 💾 **Save As** button next to "Output Excel File"
2. Choose a location and filename for the Excel output
3. The file will automatically have the `.xlsx` extension

Step 3: Select Artifacts to Process

Use the checkboxes to select which artifacts you want to extract. You can:

- **Select All** - Click "✓ Select All" to enable all artifacts
- **Deselect All** - Click "✗ Deselect All" to clear all selections
- **Custom Selection** - Manually check/uncheck individual artifacts

Available Artifacts:

Category	Artifact	Description
History	History	Complete browsing history
	History Gaps	Gaps in history record numbering (potential deletions)
Downloads	Downloads	Downloaded file information
	Downloads Gaps	Gaps in download record numbering
Form Data	Autofill	Form field autofill entries
	Addresses	Saved address information
	Credit Cards	Masked credit card information
	Bank Accounts	Masked bank account information with usage data
	Loyalty Cards	Store loyalty/rewards card information
Login	Login Data	Saved login credentials (passwords encrypted)
	Login Data Gaps	Gaps in login record numbering
Search	Search Terms	Search queries and associated URLs
	Keywords	Search engine keyword information
Navigation	Shortcuts	Omnibox shortcuts and visit counts
	Top Sites	Most visited websites
Bookmarks	Bookmarks	Saved bookmarks (including backup)
Cookies	Cookies	Cookie data and timestamps
Media	FavIcons	Website favicons and metadata
Settings	Preferences	User preferences and account information
Edge Only	Web Assist (Edge)	Edge-specific navigation history

Step 4: Run the Parser

1. Click the 🚀 **Run Parser** button
2. Monitor progress in the status window

3. Wait for the "Processing completed successfully!" message

Step 5: Access Results

When processing completes:

- A success dialog will display the output file location
- Open the Excel file to review the extracted data

Stopping Processing

To stop processing before completion:

1. Click the ■ **Stop** button
2. Partial results will be available in the output file

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6. Understanding the Output

Excel File Structure

The output Excel file contains multiple worksheets:

1. **Summary** (first sheet) - Record counts for each artifact category
2. **Preferences** (second sheet) - Parsed user preferences and account info
3. **Individual artifact sheets** - One sheet per processed artifact

Summary Worksheet

The Summary worksheet provides a quick overview of extracted data:

Worksheet Name	Record Count
History	15,432
Downloads	287
Cookies	3,891
...	...

Timestamp Columns

Most worksheets include both raw and decoded timestamp columns:

- **Raw timestamps** - Original WebKit format (microseconds since January 1, 1601)
- **Decoded timestamps** - Human-readable UTC format (e.g., "2024-03-15 14:32:45")

Key Columns by Artifact

History:

- `urls.id` - Unique identifier for the URL
- `url` - The visited URL
- `title` - Page title
- `visit_count` - Number of visits
- `typed_count` - Number of times URL was typed
- `Visit Source` - Local, Synced, Extension, or Imported
- `Transition Type` - How the user navigated to the URL
- `Decoded visit_time (UTC)` - When the visit occurred

Downloads:

- `target_path` - Where the file was saved
- `referrer` - Page that initiated the download
- `Decoded start_time (UTC)` - When download started
- `Decoded danger_type` - Security assessment
- `Download Source` - URL of the downloaded file

Login Data:

- `signon_realm` - Domain for the login
- `username_value` - Saved username
- `Decoded date_created (UTC)` - When credential was saved
- `times_used` - Number of times used

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7. Supported Browsers

Google Chrome

Full support for all artifacts. This is the primary browser the tool was developed and tested against.

Microsoft Edge

Full support with additional Edge-specific features:

- Automatically detects Edge profiles by checking for "edge" in the path
- Parses the `WebAssistDatabase` unique to Edge
- Extracts Edge-specific account information from Preferences

To parse Edge profiles:

1. Ensure "edge" appears somewhere in the profile path
2. Enable the "Web Assist (Edge)" artifact checkbox

Opera

Tested and working with Opera 113. Same artifact support as Chrome.

Other Chromium Browsers

The tool should work with other Chromium-based browsers including:

- Brave
- Vivaldi
- Chromium

However, these may have unique database tables or fields that are not parsed. Results may vary.

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8. Artifact Reference

History Artifacts

History Database (History file)

The History SQLite database contains browsing history with detailed navigation information:

Field	Description
Transition Types	Link clicked, URL typed, Suggestion clicked, Form submitted, etc.
Qualifiers	Forward/Back button, From address bar, Home page, External app
Visit Duration	Time spent on the page in seconds
Segments	URL path segments for organization

History Gaps

Detects deleted history records by finding gaps in record ID sequences. Results include:

- Previous and current record numbers
- Number of missing records
- Timestamp boundaries of the gap

Download Artifacts

Downloads

Extracted from the `downloads` and `downloads_url_chains` tables:

Field	Description
State	Complete or Interrupted
Danger Type	Security classification (0=Safe, 1-23=Various risks)
Interrupt Reason	Why download stopped (network, user, server issues)

Web Data Artifacts

Autofill

Form field data users have entered, including:

- Field name and value
- Usage count
- Creation and last used dates

Addresses

Saved address book entries with:

- Name, email, phone
- Street address, city, state, postal code, country
- Office/company information

Credit Cards

Masked credit card information (last 4 digits only):

- Card network (Visa, Mastercard, etc.)

- Expiration date
- Billing address reference

Bank Accounts

Masked bank account information including:

- Bank name and account type
- Account number suffix (last digits only)
- Account nickname
- Usage count and last use date

Loyalty Cards

Store loyalty and rewards card information including:

- Merchant name and domain
- Program name and logo URL
- Loyalty card number

Preferences Artifact

Parsed from the JSON Preferences file:

Information	Description
Account Info	Email, full name, GAIA ID, profile picture URL
Profile	Creation date, browser version, profile name
Settings	Download directory, startup pages, homepage
Exit Status	Normal or Crashed
New Tab Shortcuts	Custom links on new tab page

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9. Troubleshooting

Common Issues

"Database is locked" Error

Cause: The browser is still running and has locked the database files.

Solution:

1. Close all browser windows completely
2. Check Task Manager for any remaining browser processes
3. Try again

"No such file or directory" Error

Cause: The selected folder is not a valid browser profile or is missing expected files.

Solution:

1. Verify you selected the correct profile folder (should contain files like `History`, `Cookies`, `Preferences`)
2. Ensure you're selecting the profile folder, not the parent "User Data" folder

Empty or Missing Worksheets

Cause: The corresponding database table may be empty or the browser version uses a different structure.

Solution:

1. Check the Summary worksheet for "0 records" entries
2. Some artifacts (like Bank Accounts) may legitimately be empty
3. Older browser versions may use different table structures

"Module not found" Error

Cause: Required Python packages are not installed.

Solution:

```bash
pip install -r requirements.txt
```

Performance Issues

Processing Takes Too Long

For profiles with extensive history:

1. Select only the artifacts you need
2. Consider processing during off-hours
3. Ensure adequate disk space for the output file

Large Output Files

History-heavy profiles can generate large Excel files:

1. Excel may be slow to open very large files
2. Consider filtering data in Excel after opening
3. Use Excel's filter and sort features to find relevant data

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10. Frequently Asked Questions

Q: Can this tool decrypt saved passwords?

A: No. The tool extracts encrypted password data, but decryption requires the user's Windows credentials and is not implemented. The `password_value` field will contain encrypted binary data.

Q: Does this tool modify the original browser data?

A: No. The tool only reads data; it never writes to or modifies the browser profile. For forensic integrity, always work with a copy of the profile.

Q: Can I use this on a live system while the user is browsing?

A: Not recommended. Browser databases are locked during use, causing errors. Always close the browser first.

Q: What's the difference between "History" and "History Gaps"?

A: "History" extracts all browsing records. "History Gaps" identifies where records have been deleted by finding missing sequence numbers in the database, which can indicate evidence of history clearing or tampering.

Q: Why is "Web Assist (Edge)" disabled by default?

A: This artifact is specific to Microsoft Edge and doesn't exist in Chrome or other browsers. Enable it only when analyzing Edge profiles.

Q: How far back does browser history go?

A: Chrome retains approximately 90 days of history by default. Bookmarked URLs may persist longer in the `urls` table even after their visit records expire.

Q: Can I analyze multiple profiles at once?

A: Currently, the tool processes one profile at a time. Run it separately for each profile and specify different output files.

Q: What timezone are the decoded timestamps in?

A: All decoded timestamps are in UTC (Coordinated Universal Time). Convert to local time as needed for your investigation.

Appendix A: File Locations Quick Reference

Browser	Windows Path
Chrome	%LOCALAPPDATA%\Google\Chrome\User Data\Default
Edge	%LOCALAPPDATA%\Microsoft\Edge\User Data\Default
Opera	%APPDATA%\Opera Software\Opera Stable
Brave	%LOCALAPPDATA%\BraveSoftware\Brave-Browser\User Data\Default
Vivaldi	%LOCALAPPDATA%\Vivaldi\User Data\Default

Appendix B: Transition Type Reference

Value	Type	Description
0	LINK	Clicked a link
1	TYPED	Typed URL in address bar
2	AUTO_BOOKMARK	Clicked suggestion in UI
3	AUTO_SUBFRAME	Automatic subframe navigation
4	MANUAL_SUBFRAME	User-initiated subframe navigation
5	GENERATED	Typed text that didn't look like URL
6	START_PAGE	Top-level navigation
7	FORM_SUBMIT	Submitted form data
8	RELOAD	Reloaded the page
9	KEYWORD	Keyword-generated URL
10	KEYWORD_GENERATED	Visit generated for keyword

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